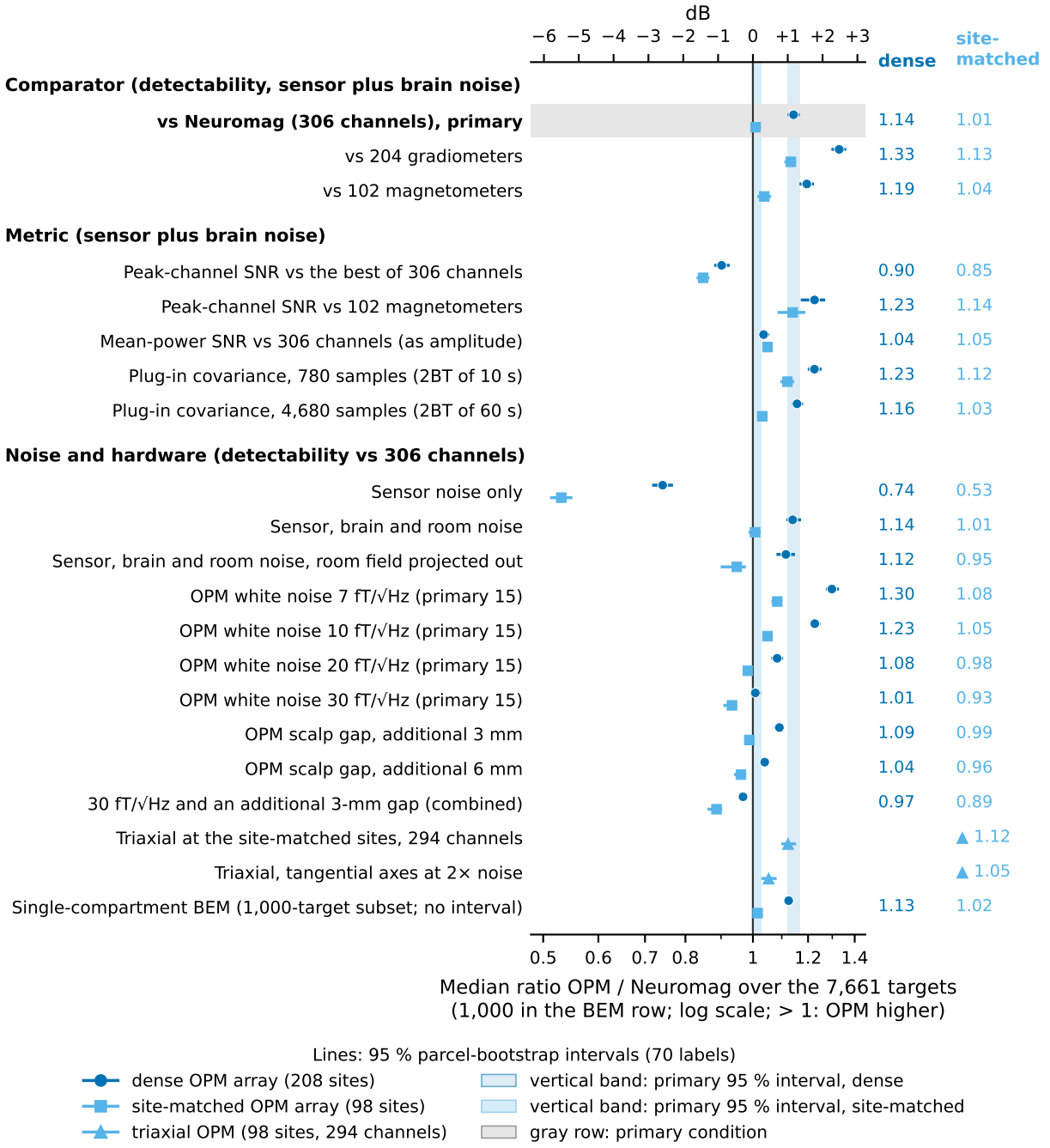
dB
−6 −5 −4 −3 −2 −1 0 +1 +2 +3
dense
site-matched
Comparator (detectability, sensor plus brain noise)
vs Neuromag (306 channels), primary
1.14
1.01
vs 204 gradiometers
1.33
1.13
vs 102 magnetometers
1.19
1.04
Metric (sensor plus brain noise)
Peak-channel SNR vs the best of 306 channels
0.90
0.85
Peak-channel SNR vs 102 magnetometers
1.23
1.14
Mean-power SNR vs 306 channels (as amplitude)
1.04
1.05
Plug-in covariance, 780 samples (2BT of 10 s)
1.23
1.12
Plug-in covariance, 4,680 samples (2BT of 60 s)
1.16
1.03
Noise and hardware (detectability vs 306 channels)
Sensor noise only
0.74
0.53
Sensor, brain and room noise
1.14
1.01
Sensor, brain and room noise, room field projected out
1.12
0.95
OPM white noise 7 fT/√Hz (primary 15)
1.30
1.08
OPM white noise 10 fT/√Hz (primary 15)
1.23
1.05
OPM white noise 20 fT/√Hz (primary 15)
1.08
0.98
OPM white noise 30 fT/√Hz (primary 15)
1.01
0.93
OPM scalp gap, additional 3 mm
1.09
0.99
OPM scalp gap, additional 6 mm
1.04
0.96
30 fT/√Hz and an additional 3-mm gap (combined)
0.97
0.89
Triaxial at the site-matched sites, 294 channels
▲ 1.12
Triaxial, tangential axes at 2× noise
▲ 1.05
Single-compartment BEM (1,000-target subset; no interval)
1.13
1.02
0.5 0.6 0.7 0.8 1 1.2 1.4
Median ratio OPM / Neuromag over the 7,661 targets
(1,000 in the BEM row; log scale; > 1: OPM higher)
Lines: 95 % parcel-bootstrap intervals (70 labels)
dense OPM array (208 sites)
site-matched OPM array (98 sites)
triaxial OPM (98 sites, 294 channels)
vertical band: primary 95 % interval, dense
vertical band: primary 95 % interval, site-matched
gray row: primary condition